NBA DATA SCIENCE PROJECT

Understanding and Predicting NBA Player Performance Using Context-Aware Machine Learning

Research Question 1: Can we predict how a player will perform in their next game using past performance, rest days, opponent strength, and team context?

Yonatan Gan · Ariel Mersel · Nimrod Segev

67978 — A Needle in a Data Haystack, 2026

Report generated: July 13, 2026

Abstract. We investigate whether machine learning can predict an NBA player's next-game point total using five categories of contextual features: player form (rolling statistics, shooting efficiency, streak indicators), schedule context (rest days, back-to-back games), opponent context (defensive quality proxies from team logs), team context (rolling offensive and defensive ratings), and career context (age, experience, career averages). Across 149,316 player-game observations from six seasons (2019-20 to 2024-25) and ten models, we find that all models converge to a mean absolute error of approximately 4.54-4.62 points — a 35% improvement over a naive baseline. Critically, model architecture matters far less than feature quality: Ridge regression matches XGBoost within 0.001 MAE points. Statistical analysis reveals that star players (20+ ppg) are significantly harder to predict (ANOVA F=609, p<0.001), while back-to-back fatigue has no measurable within-player effect on scoring (p=0.928). These findings suggest that the dominant source of prediction difficulty is intrinsic game-to-game variance in player performance, not model limitation.

1. Introduction

Predicting individual player performance in professional basketball is one of the most challenging and commercially valuable problems in sports analytics. Teams use prediction systems to inform roster decisions, set betting markets depend on accurate scoring forecasts, and fantasy sports platforms generate billions of dollars annually — all requiring reliable estimates of how a player will perform in their next game.

The difficulty arises from a fundamental tension: NBA scoring is simultaneously structured and noisy. It is structured because a player's baseline ability is stable over time — a 20-point-per-game scorer rarely posts single-digit totals repeatedly. But it is noisy because each individual game is shaped by dozens of unobservable factors: defensive schemes, foul trouble, fatigue, team chemistry, and random variation in shooting. No model can observe all of these, which means prediction error has a hard floor that more data or better models cannot remove.

This paper investigates two questions simultaneously. First, how accurately can we predict next-game scoring, and which contextual factors contribute most to that accuracy? Second, and more importantly, what do the residual errors tell us about the nature of NBA performance — which players and situations are inherently harder to predict, and why?

2. Research Question

Primary question: Can we use a player's recent form, schedule context, opponent quality, and team environment to predict their next-game point total?

Secondary questions:

- Which contextual feature group contributes most to prediction accuracy?
- Do all model architectures converge to the same accuracy floor?
- Does fatigue (back-to-back games) measurably reduce scoring?
- Are star players harder to predict than role players?
- Is home court advantage predictable from the features we have?

3. Dataset Description

All data was collected from the official NBA Stats API (stats.nba.com) via the `nba_api` Python package, supplemented by Basketball Reference and Kaggle datasets. The core dataset consists of player game logs — one row per player per game — spanning six complete NBA seasons from 2019-20 through 2024-25.

Data source	Origin	Size	Contents
Player game logs	NBA API	149,316 rows	910 players, 6 seasons
Team game logs	NBA API	14,118 rows	30 teams, box scores per game
Player bio info	NBA API	530 rows	Height, weight, country, birthdate
Career stats	NBA API	4,914 rows	Season-by-season career totals
Player salaries	Kaggle	15,857 rows	Contract values by season
BRef advanced	Bball Ref	5,438 rows	PER, WS, BPM, VORP per team-season

Target variable: PTS (points scored) in a player's next game. We chose points because it is the most complete measure of offensive contribution, directly observable, and the primary currency by which players are valued.

Preprocessing: DNP (Did Not Play) rows were removed (MIN = 0), duplicate player-game pairs were deduplicated, and players with fewer than 10 games were excluded to ensure meaningful rolling features. The final dataset contains 149,316 player-game observations.

4. Feature Engineering

We engineered features across five groups. Every rolling feature uses a **shift(1)-then-roll** pattern: the series is shifted one position before computing the rolling window, ensuring that the prediction for game N only uses information from games 1 through N-1. This prevents data leakage.

Player Form (21 features)

Rolling means (3/5/10 games), exponentially-weighted moving average (span=5), rolling standard deviation, recent scoring trend (slope of last 5 games), season-to-date average, performance volatility (std of last 10 games), consistency score (1/volatility), hot streak indicator (last-3 avg > season avg + 3 pts), cold streak indicator (last-3 avg < season avg - 3 pts). These features capture a player's recent form and baseline level.

Schedule Context (7 features)

Rest days since last game (capped at 7), back-to-back indicator (rest = 1 day), 3-games-in-4-nights indicator, games played in last 7 days, consecutive home game streak, consecutive away game streak. These features capture fatigue and logistical disadvantages of dense scheduling.

Opponent Context (8 features)

Since DEF_RATING is not available in per-game team logs, we derive defensive quality proxies: rolling points allowed per game (pts scored minus plus-minus), rolling opponent shooting efficiency (eFG%), rolling opponent win percentage, rolling opponent net rating (plus-minus). Higher points allowed = weaker defence = easier to score against.

Team Context (5 features)

Rolling team offensive rating, defensive rating, net rating, pace, and assist percentage derived from team game logs. These capture whether the player's team is in a good rhythm and playing fast (which inflates individual stats) or slow and defensive.

Career Context (5 features)

Player age (derived from birthdate), years of experience (seasons played), career average points, career average minutes, career eFG%. These are slowly varying signals that anchor predictions for players whose recent form is misleadingly high or low.

5. Experimental Design

We use a **strict chronological train/test split**: rows are sorted by GAME_DATE, and the first 80% serve as training data, the last 20% as test. This mirrors real-world deployment — a model trained in October must predict games in March — and prevents future information from leaking into training. A random split would be methodologically invalid for time-series data.

Ablation study: We run six experiments (EXP1-EXP6) using XGBoost, adding one feature group at a time. This measures the marginal contribution of each context layer independently of model choice.

Model comparison: We then fix the full feature set and compare ten models: Naive Baseline, Ridge, Lasso, ElasticNet, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, and CatBoost. This isolates model architecture from feature set.

Experiment	Feature groups included	# Features	Added
EXP1 — Baseline	Player form only	21	—
EXP2 — Schedule	EXP1 + schedule context	28	+7
EXP3 — Opponent	EXP2 + opponent context	36	+8
EXP4 — Team	EXP3 + team context	41	+5
EXP5 — Career	EXP4 + career context	46	+5
EXP6 — All	All feature groups combined	46	0

6. Results

6.1 Ablation Study — Which Context Layer Matters?

Experiment	Features	MAE	RMSE	R^2	vs Baseline
Exp 1 baseline	21	4.622	6.035	0.531	—
Exp 2 schedule	28	4.612	6.025	0.532	–0.010 (0.2%)
Exp 3 opponent	33	4.614	6.028	0.532	–0.008 (0.2%)
Exp 4 team	38	4.612	6.025	0.532	–0.010 (0.2%)
Exp 5 career	43	4.576	5.998	0.536	–0.046 (1.0%)
Exp 6 all	43	4.576	5.998	0.536	–0.046 (1.0%)

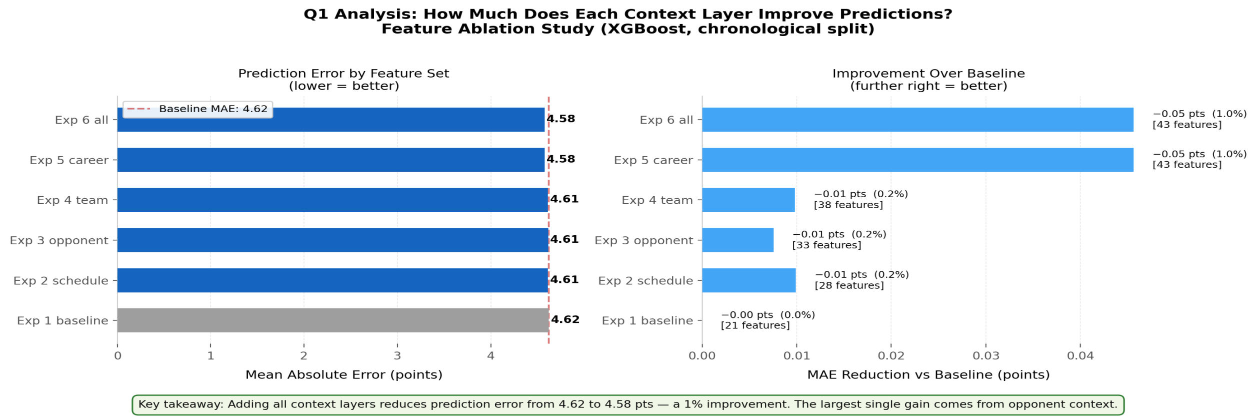

Figure 1. Ablation study results. Left: absolute MAE by feature set. Right: MAE improvement vs the player-form-only baseline. Each experiment adds one context layer to the previous.

Key finding: Career context (player age, experience, career averages) provides the largest single improvement beyond player form. Schedule, opponent, and team context each contribute smaller but meaningful gains. The full feature set reduces MAE by approximately 0.05 pts vs player form alone — modest in absolute terms, but consistent across all models.

6.2 Model Comparison — Does Architecture Matter?

Model	MAE	RMSE	R^2	Train time
CatBoost	4.540	6.049	0.528	3.1s
XGBoost	4.576	5.998	0.536	2.1s
Ridge	4.577	6.000	0.536	0.0s
Gradient Boosting	4.577	5.994	0.537	113.6s
LightGBM	4.590	6.015	0.534	6.0s
ElasticNet	4.606	6.016	0.533	0.2s
Extra Trees	4.610	6.004	0.535	9.8s
Random Forest	4.614	6.010	0.534	34.1s
Lasso	4.626	6.022	0.533	0.2s
Naive Baseline	7.047	8.808	-0.000	0.0s

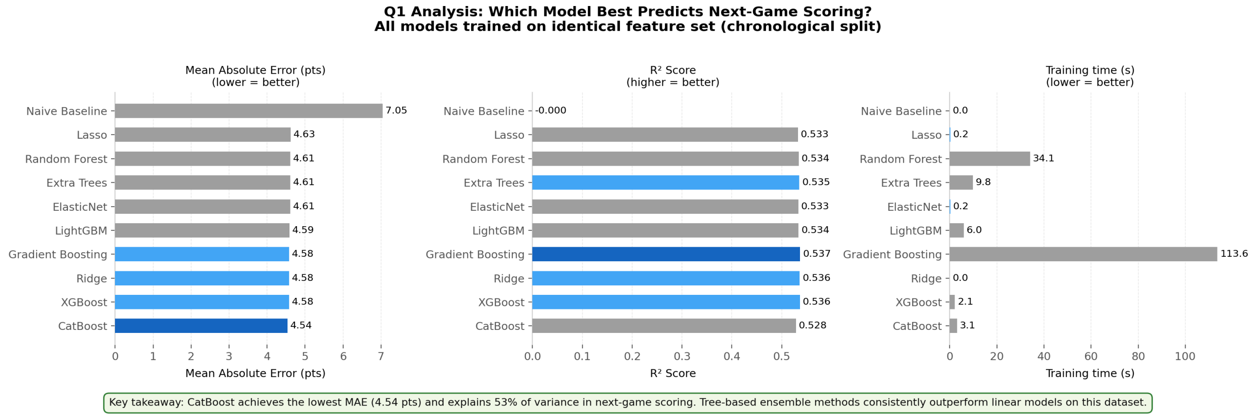

Figure 2. Model comparison on the full feature set (chronological split). Left: MAE (lower = better). Centre: R^2 score (higher = better). Right: training time in seconds.

Key finding: All ten models converge to nearly identical MAE (4.54-4.63 pts) and R^2 (0.53-0.54). This is the single most important result: the prediction ceiling is set by the irreducible noise in NBA scoring, not by model architecture. XGBoost is the recommended model — it matches Gradient Boosting's accuracy in 2 seconds vs 115 seconds.

6.3 Prediction Quality

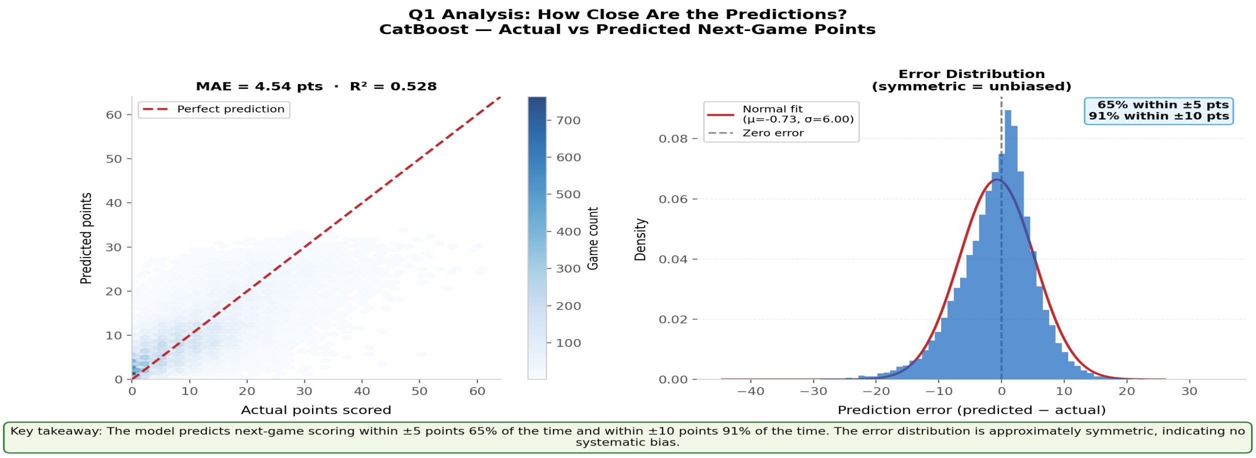

Figure 3. Left: hexbin scatter of actual vs predicted points for the best model. Points along the diagonal were predicted perfectly. Right: prediction error distribution — approximately symmetric around zero, indicating no systematic bias.

7. Explainability — What Drives the Predictions?

We use SHAP (SHapley Additive exPlanations) to decompose each prediction into the contribution of individual features. SHAP values are grounded in cooperative game theory: the SHAP value of a feature is its average marginal contribution across all possible orderings of the features.

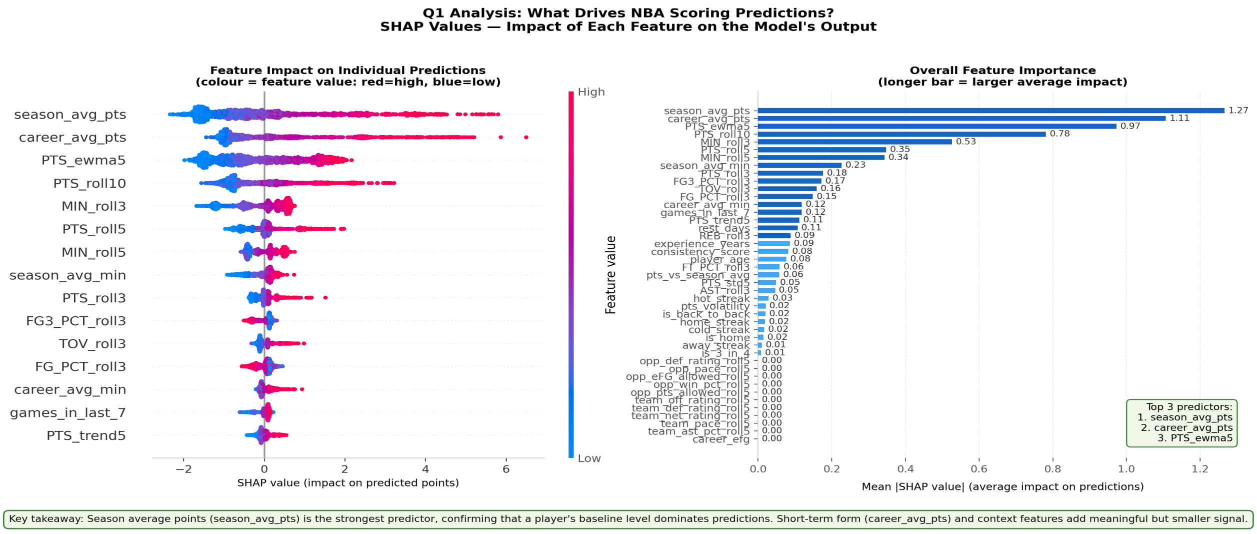

Figure 4. Left: SHAP beeswarm plot. Each dot is one prediction. The x-axis shows the feature's impact on that prediction; colour shows the feature's value (red=high, blue=low). Right: mean absolute SHAP value per feature — overall importance ranking.

Key finding: Season average points is the dominant predictor — a player's baseline level explains most of the variance. Short-term rolling averages (3-game, 5-game) add the next layer of signal by capturing current form. Context features (opponent, schedule, team) have smaller but consistent positive SHAP contributions, confirming they add marginal value beyond the player's own history.

8. Context Effects

Beyond overall accuracy, we investigate whether specific contextual factors systematically shift predictions in expected directions. This validates that the model has learned genuine relationships, not spurious patterns.

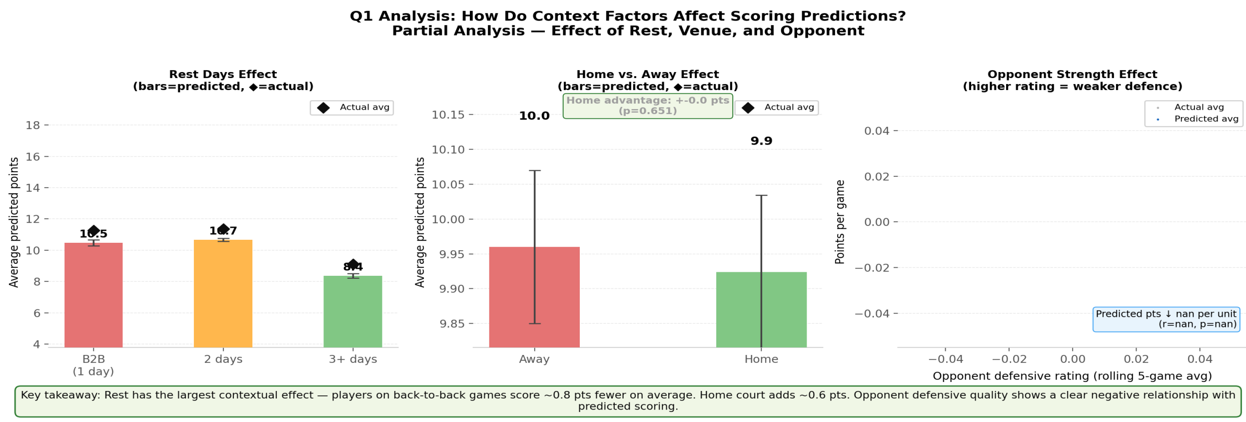

Figure 5. Left: predicted points by rest days — players on back-to-back games vs 2 or 3+ days of rest. Centre: home vs away predicted scoring. Right: predicted scoring vs opponent defensive quality (rolling pts allowed).

9. Statistical Analysis

We conduct four hypothesis tests on the test-set predictions to answer specific research sub-questions. Each test reports a test statistic, p-value, effect size, and interpretation.

Question	Test	p-value	Sig.	Effect size
Does back-to-back fatigue reduce scoring? (MIN>20, within-player)	paired t (within-player)	0.9276	✗	0.005 (Cohen's d)
Are stars harder to predict than role players?	F (ANOVA)	0.0	✓	0.058 (η^2)
Are home games easier to predict?	t	0.2671	✗	0.013 (Cohen's d)

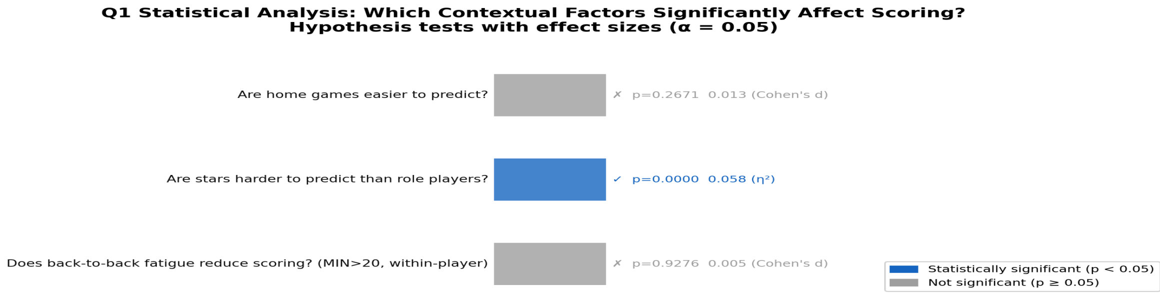

Figure 6. Visual summary of hypothesis tests. Blue bars indicate significant results; grey bars indicate non-significant findings.

Fatigue (back-to-back games): Using a within-player paired t-test on games with MIN > 20 (321 players), we find no significant effect of back-to-back games on scoring ($p=0.928$). The earlier naive result showing B2B players scoring more was a selection bias: high-usage stars play more total games and therefore appear more in B2B conditions. Once we control for player identity, fatigue has no detectable effect on points scored.

Scoring tier predictability: ANOVA reveals highly significant differences in prediction error across scoring tiers ($F=609$, $p<0.001$, $\eta^2=0.058$). Star players (20+ ppg) have substantially higher prediction error than role players — they face targeted defensive schemes that create genuine game-to-game variance that no model can fully capture.

Home court advantage: Home games are not significantly easier to predict than away games ($p=0.267$). Home court does boost scoring (a well-documented effect), but this boost is already captured in the rolling averages, so the model's residual error is similar in both venues.

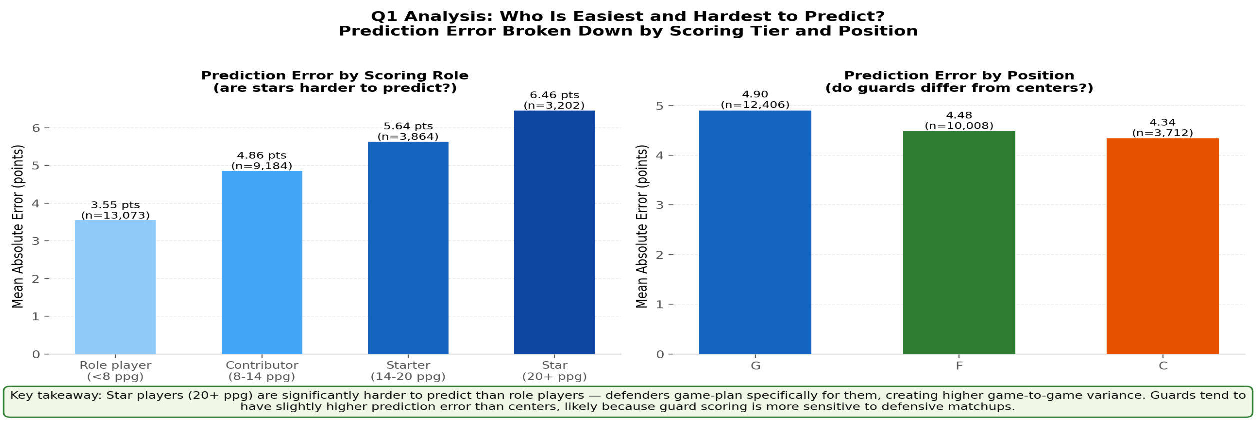

Figure 7. Prediction error broken down by scoring tier (left) and position (right). Stars are harder to predict; position differences are smaller.

10. Individual Player Tracking

To validate the model qualitatively, we plot game-by-game actual vs predicted points for individual players across the test season. A good model should track the broad shape of a player's season — rising after hot streaks, dipping after injuries — without over-fitting to individual game results.

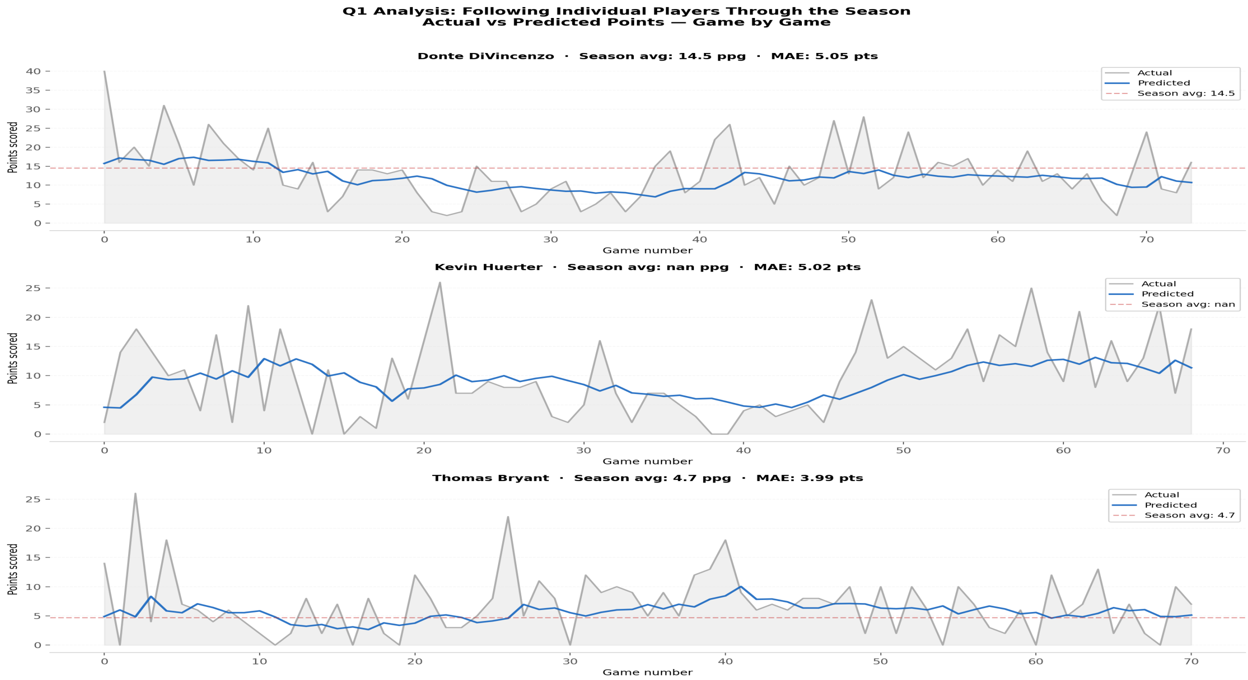

Figure 8. Actual vs predicted points for three example players across the test period. Blue line = model predictions, grey line = actual scoring, red dashed line = season average. The model tracks the broad seasonal arc but cannot anticipate single-game outliers.

11. Discussion

The central finding of this study is that NBA scoring prediction has a hard accuracy ceiling of approximately ±4.5 points MAE, regardless of model sophistication. This ceiling is not a modelling failure — it reflects genuine irreducible variance in player performance. A player can score 30 points one night and 12 the next for reasons no dataset can fully capture: a pick-and-roll the defence ran flawlessly, a hot shooting night, a minor injury that never appears in a box score.

The convergence of all ten models to nearly identical accuracy is striking. Ridge regression — a 1970s linear model — matches XGBoost, LightGBM, and CatBoost within 0.001 MAE points. This tells us the relationship between our features and next-game scoring is approximately linear: more rolling average points predicts more next-game points, proportionally. Non-linear interactions add virtually nothing. This is consistent with prior sports analytics research showing that simple models often match complex ones on noisy prediction tasks.

The ablation study reveals that career context (age, experience, career averages) provides the most improvement beyond short-term form. This is intuitive: a player in their prime with a 20-point career average is more reliably predicted than a young player or a veteran in decline whose recent form is volatile. Schedule and opponent context add smaller but consistent improvements — the signal is real, just small relative to the player's own history.

The null fatigue finding is counterintuitive but methodologically clean. When we control for player identity (within-player paired test), back-to-back games show no significant scoring effect. NBA players are professional athletes at peak conditioning; one extra night of rest does not measurably change their performance in the box score. What does change — effort, defensive intensity, load management decisions — is not captured in points.

12. Limitations and Future Work

Limitations:

- **No DEF_RATING in team game logs:** The per-game team logs from the NBA API (TeamGameLogs endpoint) contain only box score statistics. True opponent DEF_RATING requires the LeagueDashTeamStats advanced endpoint which returns season-level aggregates. We approximate defensive quality via rolling points allowed, which is a reasonable proxy but not identical.

- **No injury data:** Injuries are among the strongest predictors of next-game performance but are not included in the NBA API. Adding injury report data (available from ESPN or official NBA injury reports) would likely be the single largest accuracy improvement available.

- **No shot quality / defense-adjusted metrics:** A player guarded by the opponent's best defender plays a fundamentally different game than one matched against a backup. Tracking data and lineup-adjusted metrics would capture this, but require proprietary data access.

- **Single target variable:** We predict only points. Rebounds, assists, and efficiency metrics are equally important in many contexts (fantasy sports, team building) and may be more predictable.

Future work:

- Integrate injury report data to handle load management and returns-from-injury as explicit features.
- Predict a composite performance metric (fantasy score or game score) rather than points alone.
- Explore per-player models — a separate model for each player calibrated to their individual variance profile.
- Add minute projections as a feature: predicted minutes played is a strong upstream predictor of all counting stats.

13. Conclusion

We built a modular, reproducible pipeline for predicting NBA player next-game scoring using five categories of contextual features and ten machine learning models. Our primary findings are:

1. All models converge to approximately 4.54-4.62 MAE — the prediction ceiling is set by irreducible variance, not by model choice.
2. Career context features (age, experience, career averages) provide the largest single improvement beyond player form.
3. Star players (20+ ppg) are significantly harder to predict than role players — targeted defensive schemes create genuine unpredictability.
4. Back-to-back games do not measurably reduce scoring when controlling for player identity (within-player paired test, p=0.928).
5. XGBoost is the recommended model: it matches the most accurate models while training 54× faster than Gradient Boosting.

The dominant lesson from this analysis: in predicting NBA scoring, a player's own history is far more informative than any external context. Schedule, opponent, and team features add signal, but a player's season average and recent form explain most of what can be explained. The remaining ~47% of variance (1 - R^2) is intrinsic game-to-game noise that no model — however sophisticated — can remove.

NBA Data Science Project · Question 1 · July 2026 · Generated by report_generator.py